



FINDING FRAMEWORKS AND BENCHMARKS

GRAD 695



FRAMEWORKS

- Good research is contextualized within the history of what has been done
- For different types of problems, frameworks exist for the best (or at least common) approaches
- Types of Frameworks:
 - Analytic such as Predictive Modeling or Forecasting
 - Theoretical such as Game Theory or Information Theory
 - Methodological such as Experimental or Data Mining
- May incorporate more than one framework

BENCHMARKS

- For the majority of analytic problems, some model or solution already exists
- New models or solutions must be compared to them
- As you define your problem and methodological approach, establish what the baseline/best performance is for that task or problem
- Types of Benchmarks:
 - Benchmark Datasets: established datasets meant to test new models or solutions to common tasks (e.g. ImageNet is a common benchmark for basic image classification)
 - Benchmark Metrics: recorded metrics for current best performance for a given task under given conditions
- Exploratory or Descriptive research generally don't have concrete benchmarks, but should be compared to what is currently known about the topic or area

- To determine the appropriate frameworks and benchmarks, conduct a literature search
- Ways to find literature:
 - Google Scholar
 - Ask ChatGPT (but verify the sources)
 - Reference sections
- Sources should be peer-reviewed/scholarly sources
 - White papers are fine, but blog posts, news articles, or other informal sources are not
- Once you have 10-20 sources, you can begin writing the literature review (or related works) section of your proposal

ESTABLISHING FRAMEWORKS AND BENCHMARKS

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PURPOSE OF A LITERATURE REVIEW

- ▶ Relate your work to ongoing discussion in the field
- ▶ Framework for establishing the importance of the RQ
- ▶ Benchmark for Comparing Results and Conclusions
 - ▶ How much better is your model?
 - ▶ How much more variance can you explain in X?
 - ▶ What have you added to what is known?

WHAT A LITERATURE REVIEW IS

- ▶ A logical argument based on a comprehensive understanding of state of research on your topic
- ▶ All claims are supported by credible sources
- ▶ Shows how your proposed project relates to past work
 - ▶ Very important to explicitly discuss how your proposed work extends on past studies or addresses limitations in past work
- ▶ Must be organized and have a logical flow from your goal to your hypotheses

WHAT A LITERATURE REVIEW IS NOT

- ▶ A series of summaries for your sources
- ▶ A annotated bibliography
- ▶ A data dump of everything you can find

- ▶ Zotero, <https://www.zotero.org/>, is a free, open-source citation management tool
- ▶ Highly recommend using to manage and generate citations and to keep notes and records
- ▶ HU's Library has put together [resources and guides](#) to help you use it

CITATION MANAGEMENT



Literature Map

Subheadings

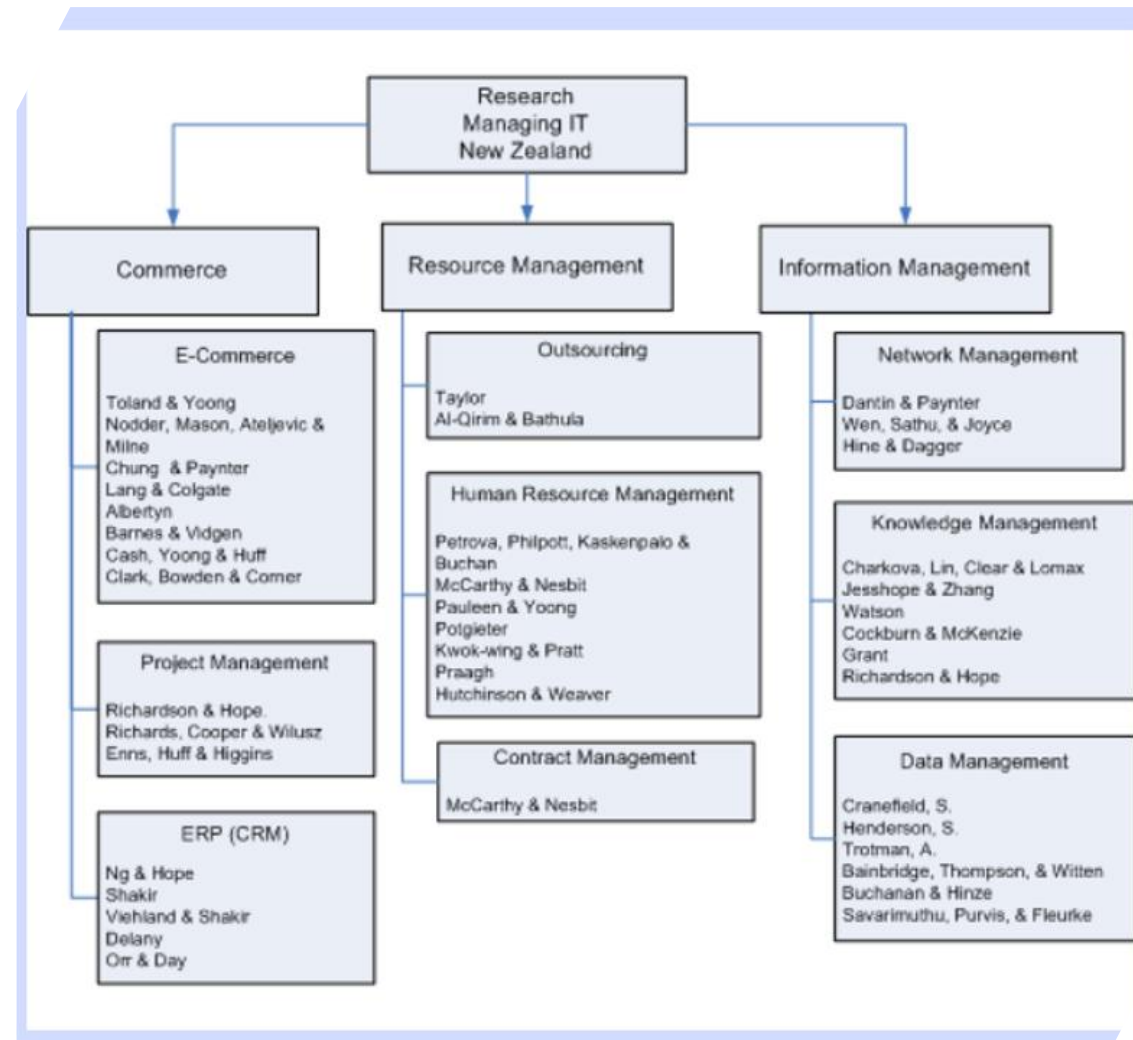
First Sentences

METHODS FOR
ORGANIZING
AND
OUTLINING A
LITERATURE
REVIEW

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METHOD 1: LITERATURE MAPS

- ▶ Literature maps: a diagram linking important concepts/constructs and sorting sources by these concepts
- ▶ Plenty of examples and templates online, [Examples](#)



Basically a literature map but formatted as a bulleted list

Should cover all the necessary concepts and theories relevant to your project goal

Start from broadest concepts and oldest, most traditional methods/approaches before narrowing the focus and moving to what is unknown/evolving

Generally, should have no more than 2 subheadings per page of text and at least 4 citations under each subheading

METHOD 2: SUBHEADINGS

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- ▶ Example Topic: The influence of Twitter marketing on tech stocks
- ▶ Example Subheadings:
 - ▶ Twitter marketing
 - ▶ Taecharungroj, V. (2017). Starbucks' marketing communications strategy on Twitter. *Journal of Marketing Communications*, 23(6), 552-571.
 - ▶ Leung, X. Y., Bai, B., & Stahura, K. A. (2015). The marketing effectiveness of social media in the hotel industry: A comparison of Facebook and Twitter. *Journal of Hospitality & Tourism Research*, 39(2), 147-169.
 - ▶ Tech industry marketing
 - ▶ Nouri, P., & Ahmady, A. (2018). A taxonomy of nascent entrepreneurs' marketing decisions in high-tech small businesses. *Journal of Small Business Strategy*, 28(3), 69-79.
 - ▶ Yang, M., & Gabrielsson, P. (2017). Entrepreneurial marketing of international high-tech business-to-business new ventures: A decision-making process perspective. *Industrial Marketing Management*, 64, 147-160.
 - ▶ Factors impacting tech stocks
 - ▶ Bento, N., Gianfrate, G., & Groppo, S. V. (2019). Do crowdfunding returns reward risk? Evidences from clean-tech projects. *Technological Forecasting and Social Change*, 141, 107-116.
 - ▶ Lee, C. C., Chen, M. P., & Chang, C. H. (2013). Dynamic relationships between industry returns and stock market returns. *The North American Journal of Economics and Finance*, 26, 119-144.



- ▶ A method for constructing an outline for your literature review as a logical argument connecting existing work on your topic to your specific approach to the problem
- ▶ Rather than just outline the concepts you need to discuss, this method forces you to think about the assumptions you are making and the premises you need to support for your research questions/hypotheses to make sense
- ▶ Can lead to a much more coherent and logical flow that is concise and to-the-point

METHOD 3: FIRST SENTENCES

EXAMPLE TOPIC: CHANGES IN THE COMMUNICATION STYLES OF POLITICAL LEADERS

- ▶ Example Outline:
 - ▶ Most scholars and political commentators on both the left and the right agree that Donald Trump is unlike any previous US president.
 - ▶ During his campaign and in the first years of his presidency, many have claimed that Trump's simplicity and directness were keys to his popularity.
 - ▶ A second major distinguishing feature of Trump's language is his self-confidence.
 - ▶ We build on an extensive research tradition of exploring political leadership traits.
 - ▶ If conveying simple messages with confidence is a powerful way to persuade voters in a democracy, it is incumbent on researchers to determine effective ways to measure these dimensions reliably, quickly, and efficiently.
 - ▶ A common distinction in language is between content and function words.
 - ▶ The eight categories of function words are not independent and indeed, capture an underlying psychological dimension known as analytic thinking.
 - ▶ Past work has validated the use of function words as markers of the analytic thinking dimension, including a study of college admission essays from over 25,000 incoming students.
 - ▶ The analysis of function words has also been useful in identifying people's relative status or "clout" in a social hierarchy.
 - ▶ Relevant to this project, the computerized text analysis program Linguistic Inquiry and Word Count (LIWC) was updated in 2015 to include empirically established measures of both analytic thinking and confidence using algorithms based on the projects of Pennebaker et al. and Kacewicz et al.
 - ▶ The algorithm developed from the project of Kacewicz et al. is formally referred to as clout.
 - ▶ The purpose of this project was to apply linguistic analyses to large corpora of texts from the American presidency, non-US leaders, legislative bodies, and cultural contexts spanning multiple decades.



- ▶ Read critically
- ▶ Write short summaries or notes about your sources
- ▶ Keep your goal in mind
- ▶ Try to find multiple citations that support your main points
- ▶ Focus on the big picture not the small details
- ▶ Unless you are developing a new model or algorithm, you don't need to extensively explain the technical components; just briefly discuss why the chosen model makes the most sense for your particular problem

FINAL GUIDANCE